

ADVANCED TOPICS IN NETWORK SCIENCE

Fall 2026

Instructor:	Sadamori Kojaku	Time:	TuTh 9:45 – 11:15
Email:	skojaku@binghamton.edu	Place:	J01 Engineering Bldg.
Office:	J19 Engineering Bldg.	Credit:	3
Office Hour:	13:00 - 15:00 Friday	Zoom:	my/skojaku.zoom

Course Pages:

1. Course Website <https://skojaku.github.io/adv-net-sci>
2. Course Github <http://github.com/skojaku/adv-net-sci>

Office Hours: Friday 13:00 - 15:00.

Main References: Below is a curated list of essential books that will be referenced during the course.

- Mark Newman. *Networks (Second Edition)*. Oxford University Press, 2018.
- Filippo Menczer, Santo Fortunato, and Clayton A. Davis. *A First Course in Network Science*. Cambridge University Press, 2020.
- James Bagrow and Yong-Yeol Ahn. *Working with Network Data: A Data Science Perspective*. Cambridge University Press, 2024.

Objectives: Networks are all around us, from the vast expanse of the Internet to the intricate web of social connections that we build in our daily lives. But networks aren't just limited to the human realm—they can be found in every corner of the natural world, such as the complex interactions between animals, proteins, viruses, and DNAs. In recent years, we have witnessed remarkable advances in AI/ML and an ever-increasing volume and quality of data. Together, they offer an unprecedented opportunity to unlock the secrets of the world around us. This course is an introduction to network data analysis from the bottom up: through the interactions with network data and tools, we will learn how to store, manipulate, compute, and leverage network data in practice, as well as their underlying theoretical foundations.

Expected Student Learning Outcomes: After completing this course, students will:

- be able to interpret and evaluate modern network science literature, concepts, methodologies, tools, and recent research topics,
- be able to conduct advanced network modeling, analysis and simulation using appropriate mathematical/computational means,
- be able to design and conduct original research using network science methods and tools, and
- be able to demonstrate integration of Systems Science and/or Industrial and Systems Engineering knowledge and techniques and advanced network modeling/analysis/simulation skills in the form of a final project.

Prerequisites: SSIE-523. Fluency in Python. Basic understanding of mathematics and statistics.

Course Outline: (*MP = Mini Project: a hands-on group session*)**August 2026:**

08/18 (Tue) Seven Bridges of Königsberg
 08/20 (Thu) Seven Bridges of Königsberg
 08/25 (Tue) Small-world Networks
 08/27 (Thu) Small-world Networks & Project Matchmaking

September 2026:

09/01 (Tue) Small-world Networks / MP
 09/03 (Thu) Network Robustness
 09/08 (Tue) No class (Monday classes meet)
 09/10 (Thu) Network Robustness
 09/15 (Tue) Network Robustness / MP
 09/17 (Thu) Friendship Paradox
 09/22 (Tue) Friendship Paradox
 09/24 (Thu) Friendship Paradox / MP
 09/29 (Tue) Clustering

October 2026:

10/01 (Thu) Clustering
 10/06 (Tue) Clustering
 10/08 (Thu) Clustering / MP
 10/13 (Tue) No class (Fall break)
 10/15 (Thu) No class (Fall break)
 10/20 (Tue) Centrality
 10/22 (Thu) Centrality
 10/27 (Tue) Centrality / MP
 10/29 (Thu) Random Walks

November 2026:

11/03 (Tue) Random Walks
 11/05 (Thu) Random Walks / MP
 11/10 (Tue) Graph Embedding
 11/12 (Thu) Graph Embedding
 11/17 (Tue) Graph Embedding / MP
 11/19 (Thu) Graph Neural Networks
 11/24 (Tue) Graph Neural Networks
 11/26 (Thu) No class (Thanksgiving)

December 2026:

12/01 (Tue) Graph Neural Networks / MP
 12/03 (Thu) Presentation
 12/08 (Tue) Presentation & wrap-up
 Final Examinations 12/10-12/16

Grading Policy:

- *Grading Item:* Quiz (10%), In-class Presentation (10%), Assignments (20%). Exam (30%). Project (30%).
- *Bonus points:* 30% equivalent of the total grading points.
 - 10% bonus for the best project (one team).
 - 10% for the best Network of the Week Presentation (one team).
 - 10% bonus for the best question-answer assignment (2% for each assignment: no limit on the number of recipients).

- *Credits*: 3 credits.
- *Grading*: Normal grading; A through F.

Course structure: “Don’t think! Feeeeeel” is a famous quote by Bruce Lee in the movie *Enter the Dragon*, and this is the guiding philosophy of this course. The primary goal of this course is to **feel** the concepts and tools of network science through pen-and-paper exercises and hands-on coding. The course will first cover the practical skills of network science, followed by their theoretical foundations. The course activities include:

- The class will begin with a weekly quiz on Brightspace about the lecture topics in the previous week. This quiz is intended to review the lecture topics and to test your understanding of the concepts and tools of network science.
- Biweekly assignment on Github Classroom. Most are coding assignments, but not all of them are. The assignments are intended for practicing and equipping you with the skills to analyze network data.
- A short lecture to cover the theoretical aspects.

Communications: We use Discord for quicker informal communications, Q&A, team discussions, and other casual conversations. We will send you an invitation link through Brightspace. Feel free to NOT use your full name (e.g., “Jane D.”) Announcements will be sent via Brightspace and Discord. Many course-related information will be shared on Discord. So, you will miss a lot of information if you are not on Discord.

Office Hours:

- **Sadamori Kojaku:** 13:00-15:00 Fridays at J19 Engineering Building (in person) or Zoom (<https://binghamton.zoom.us/my/skojaku.zoom>).

Important Dates:

Project Proposal #1	09/27 (Sun)
Project Presentation #2	12/03 and 12/08
Project Final Paper #3	12/06 (Sun)
Final Exam	12/10-12/16

Course Policy:

- **Attendance:** Residential students are expected to attend class in person; studies show that in-person attendance is associated with better learning outcomes (<https://journals.sagepub.com/doi/10.3102/0034654310362998>, <https://www.nature.com/articles/s41599-023-02590-1>, <https://www.educationnext.org/zooming-to-class-slows-student-learning/>). If you cannot attend, submit the absence request form (<https://forms.gle/yhzHoMSaKCRJXdGm7>) and email the instructor, preferably one day before class. **An absence not submitted through the form is not counted**, even if you emailed about it. Excused absences are limited to illness, accidents, job interviews, and conference travel. If you will miss more than two weeks for a legitimate reason, request an accommodation by email.
- **Laptop and mobile:** Please refrain from using laptops and mobile phones during class unless instructed.
- **Credit hours:** This is a 3-credit course. In addition to the scheduled lectures, students are expected to spend at least 6.5 hours per week on course-related work, such as readings, lab sessions, studying for tests and examinations, and preparing assignments.

- **Generative AI:** You may use AI tools as learning aids for understanding course materials. However, the final submitted assignment must be your own work. If parts of an assignment are produced by generative AI, indicate the generated parts and cite the tools used, following this guideline: <https://style.mla.org/citing-generative-ai/>

Try each assignment on your own first before turning to AI. Most assignments in this course can be solved by AI in seconds, but the purpose of an assignment is to build a skill, and that skill is only built by working through the problem yourself. If you understand the underlying concept, you can judge whether the AI's output is correct and use it effectively; if you do not, you will not be able to check its answers, and the same kind of problem will stop you again later.

- **Google Gemini:** Free for all students. Log in with your BU Google account at <https://gemini.google.com>.
- **chat.binghamton.edu:** A campus-hosted LLM chat service run by ITS. You can issue your own API key from the account settings and use it with coding agents (e.g., Claude Code, Cursor, Codex) at no cost.
- **OpenRouter** (<https://openrouter.ai>): Offers a free tier of open-weight LLMs (50 requests/day). Adding a one-time \$10 credit permanently raises this to 1,000 requests/day.
- **Data backup:** Back up your data and code regularly, for example with Google Drive, Dropbox, or GitHub (<https://github.com>). Loss of data or code (e.g., due to a laptop failure) is not an acceptable excuse for a delayed or missing submission.
- **Graduate Academic Consultants:** Binghamton University's Graduate Academic Consultants can help you with projects in this course. This is a free service. <https://www.binghamton.edu/grad-school/academic-support/graduate-academic-consultants/>

- **Disabilities:** Every attempt will be made to accommodate qualified students with disabilities (e.g. mental health, learning, chronic health, physical, hearing, vision, neurological, etc.). You must have established your eligibility for support services through Services for Students with Disabilities. Note that services are confidential, may take time to put into place, and are not retroactive. The office is located in the University Union, room 119. Captions and alternate media for print materials may take three or more weeks to get produced. Please contact Disability Services for Students at <https://www.binghamton.edu/ssd/index.html> or 607-777-2686 as soon as possible if accommodations are needed.
- **Bias-based incidents:** Any act of discrimination or harassment based on race, ethnicity, religious affiliation, gender, gender identity, sexual orientation, or disability can be reported at <https://www.binghamton.edu/diversity-equity-inclusion/reportbias.html> or to the Binghamton University Affirmative Action Officer at 607-777-4775. Sexual misconduct and Title IX
- **Sexual misconduct and Title IX:** Title IX and BU's Sexual Harassment Policy regard any form of sexual harassment as a violation of the standards of conduct required of all persons associated with the institution. If you have experienced sexual misconduct or know someone who has, you can ask support from the University Counseling Center at 607-777-2772 (counseling, advocacy, and advice services). It is also important that you know that Title IX and University policy require me to share any information brought to my attention about potential sexual misconduct with the campus Deputy Title IX Coordinator or BU's Title IX Coordinator. In that event, those individuals will work to ensure that appropriate measures are taken and resources are made available. Protecting student privacy is of utmost concern, and information will only be shared with those that need to know to ensure the University can respond and assist. Visit <https://www.binghamton.edu/counseling/resources/faculty/assault.html> and <https://www.binghamton.edu/services/title-ix/index.html> to learn more.
- **Mental health issues:** If you have any mental health issues, don't hesitate to contact BU's [University Counseling Center](#), which provides free counseling sessions. Also, please contact Disability Services for Students at [Services for Students with Disabilities](#) or 607-777-6893 as soon as possible if accommodations are needed.
- **Academic Integrity:** Academic integrity is fundamental to the mission of our university. All students are expected to uphold the highest standards of academic honesty in their coursework and research. Violations of academic integrity include plagiarism, cheating, unauthorized collaboration, fabrication, and misrepresentation. For full details on the Academic Honesty Code, procedures, and appeals process, please refer to [the Student Academic Honesty Code](#). If you are unsure about what constitutes academic dishonesty in any situation, ask your instructor for clarification.